

Cover Letter — *npj Complexity*

Dear Editors,

I am pleased to submit our manuscript, “**An Area-Preserving Hénon-Map Model for the Riemann Zeros: A Deterministic-Dynamics Approach with Quantum and Dissipative Solvers,**” for consideration as a research article in *npj Complexity*.

What the paper does. We ask a question at the interface of nonlinear dynamics, complex systems, and number theory: can a single low-dimensional *deterministic* dynamical system, without hand-tuned unfolding, reproduce both the global counting function of the Riemann zeros and their local level repulsion? The study is a direct follow-up to our recently **published** result establishing that the symbolic dynamics of the Logistic map at its band-merging point is isomorphic to the prime sieve, with the Hardy–Littlewood twin-prime constant emerging as a dynamical fixed point (Wang, *Research in Mathematics* 13(1), 2026). Given that peer-reviewed foundation, we take the natural Hilbert–Pólya step and analyze the *spectrum* of the associated map. Because the 1D Logistic map is dissipative and cannot host a unitary spectrum, we lift it to the 2D area-preserving Hénon map and study it near the edge of chaos with two numerically independent eigensolvers (a unitary Fourier solver and a Markovian dissipative solver).

Why *npj Complexity*. The work sits squarely in the journal’s scope — emergence, self-organization, and nonlinear/complex dynamics — and uses the complex-systems toolkit (edge-of-chaos dynamics, KAM breakup, spectral statistics, large-scale computation) to address a classical problem from a fresh, constructive angle. It also exemplifies a data-driven, “bottom-up” discovery workflow that we believe will interest the journal’s readership.

On rigor and claims. We are deliberately careful about epistemic status. The manuscript presents its results as *numerical observations and heuristic arguments, not proofs*, and includes an explicit table classifying every claim as established, published, numerical, heuristic, qualitative, or conjectural. We do not claim to have identified the Hilbert–Pólya operator or to have proven any statement about the Riemann Hypothesis. All code, data, and logs are openly available (https://github.com/maris205/riemann_henon) for full reproducibility.

Originality and prior consideration. The manuscript is original, has not been published elsewhere, and is not under consideration by another journal. The sole author has no competing interests.

Thank you for considering our submission. We would be happy to suggest suitable referees with expertise in quantum chaos, random matrix theory, and dynamical systems upon request.

Sincerely,

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